

Student Feedback Analysis with RAG

This notebook analyzes student feedback using LangChain agents with retrieval-augmented generation.

```
In [1]: # pip install -q langchain langchain-groq langchain-community sentence-transl

import os
import sys
import pandas as pd

# Ensure required packages are installed in this notebook kernel
required_packages = [
    "langchain",
    "langchain-groq",
    "langchain-community",
    "sentence-transformers",
    "chromadb",
    "pandas",
    "openpyxl",
    "python-dotenv",
    "langchain-huggingface",
    "langchain-chroma",
]

os.system(f"{sys.executable} -m pip install -q " + " ".join(required_packages))

from dotenv import load_dotenv
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_chroma import Chroma
from langchain_core.documents import Document
from langchain_groq import ChatGroq
from langchain.agents import create_tool_calling_agent, AgentExecutor
from langchain.tools import tool
from langchain_core.prompts import ChatPromptTemplate

# Load environment variables from .env
load_dotenv(override=True)

groq_api_key = os.getenv("GROQ_API_KEY")
if not groq_api_key:
    print("WARNING: GROQ_API_KEY not found in .env file. Please set it.")
else:
    print("Groq API Key found.")

huggingface_api_key = os.getenv("HUGGINGFACE_API_KEY")
if not huggingface_api_key:
    print("WARNING: HUGGINGFACE_API_KEY not found in .env file. Please set it.")
else:
    print("Huggingface API Key found.")
```

WARNING: You are using pip version 21.2.4; however, version 26.0.1 is available.

You should consider upgrading via the '/Users/madams1/Documents/FAU/boilerplate/.venv/bin/python -m pip install --upgrade pip' command.

/Users/madams1/Documents/FAU/boilerplate/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: <https://github.com/urllib3/urllib3/issues/3020>

warnings.warn(
/Users/madams1/Documents/FAU/boilerplate/.venv/lib/python3.9/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html

from .autonotebook import tqdm as notebook_tqdm

Groq API Key found.

Huggingface API Key found.

```
In [ ]: file_path = "feedback.xlsx"  
  
if os.path.exists(file_path):  
    df = pd.read_excel(file_path)  
    print("Data loaded. First 5 rows:")  
    print(df.head())  
else:  
    print(f"File not found: {file_path}. Please make sure the file is in the
```

Data loaded. First 5 rows:

	z_last_digits	student_writeup
0	436967	I've taken a few AI-related classes before. Th...
1	503497	I want to get a good overview of artificial in...
2	607951	As a front-end engineer with 8 years experienc...
3	454016	In completing this Artificial Intelligence cou...
4	569976	Through this course, I aim to develop a robust...

```
In [ ]: documents = []  
  
for index, row in df.iterrows():  
  
    content = "\n".join([f"{col}: {val}" for col, val in row.items() if pd.r  
  
    doc = Document(page_content=content, metadata={"row": index, "source": f  
    documents.append(doc)  
  
print(f"Processed {len(documents)} documents.")
```

Processed 134 documents.

```
In [ ]: embeddings = HuggingFaceEmbeddings(model_name="all-MiniLM-L6-v2")  
  
vectorstore = Chroma.from_documents(  
    documents=documents,  
    embedding=embeddings,  
    collection_name="student_feedback",  
    persist_directory="./chroma_db"  
)
```

```
print("Vector Database created.")
```

Vector Database created.

```
In [ ]: @tool
def get_all_feedback() -> str:
    """Get all student feedback to analyze topics and themes"""
    docs = vectorstore.get()['documents']
    if isinstance(docs, list) and len(docs) > 0:
        response_content = "\n\n---\n\n".join(docs[:25]) # Get first 25 docs
    else:
        # Fallback: search with broad query
        docs = vectorstore.similarity_search("learning goals interests", k=3)
        response_content = "\n\n".join([d.page_content for d in docs])
    return f"Total feedback entries: {len(docs)}\n\n{response_content}"

@tool
def search_feedback(query: str) -> str:
    """Search for specific feedback on a topic"""
    docs = vectorstore.similarity_search(query, k=10)
    response_content = "\n\n".join([d.page_content for d in docs])
    return response_content

print("Tools created.")
```

Tools created.

```
In [ ]: # Initialize the LLM
llm = ChatGroq(model_name="meta-llama/llama-4-scout-17b-16e-instruct", api_key=API_KEY)

# Initialize the tools
tools = [get_all_feedback, search_feedback]

# Create prompt
prompt = ChatPromptTemplate.from_messages([
    ("system", "You are a helpful assistant that analyzes student feedback."),
    ("placeholder", "{agent_scratchpad}"),
    ("human", "{input}"),
])

# Create and setup agent
agent = create_tool_calling_agent(llm, tools, prompt)
agent_executor = AgentExecutor(agent=agent, tools=tools, verbose=True)

print("Agent created and ready.")
```

Agent created and ready.

Query 1: Find Two Most Similar Answers

```
In [ ]: input_1 = "Using the complete student feedback data, find two most similar answers to the question: 'What are the most common learning goals?'

print("\n" + "="*80)
```

```
print("QUERY 1: Finding Most Similar Student Answers")
print("="*80)

result_1 = agent_executor.invoke({"input": input_1})
answer_1 = result_1['output']

print("\n" + "-"*80)
print("RESULT 1:")
print("-"*80)
print(answer_1)
```

```
=====
=====
QUERY 1: Finding Most Similar Student Answers
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> Entering new AgentExecutor chain...

Invoking: `get\_all\_feedback` with `{}`

Total feedback entries: 1340

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student_writeup: I've taken a few AI-related classes before. They taught me how to build models and some of the math, but the majority of the syllabus for this course covers topics I am not familiar with, such as search and knowledge representation. I'm excited to learn more about AI besides just neural networks! Although not included in the syllabus, I'd also love to touch on how GenAI works.

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I also hope to discuss more about the definitions of particular terminology and why it's defined that way. F

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student_writeup: I want to build a solid, practical foundation in AI, mastering core concepts like search, reasoning, and machine learning. My goal is to design intelligent agents and apply them to real problems, understanding the trade-offs between different methods. I'm eager to develop hands-on skills in implementing algorithms, evaluating models, and using AI tools effectively. Finally, I hope to strengthen my critical thinking and cultivate a thoughtful, ethical understanding of AI's role and impact in society.

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AI (LLM's, agents) works and how to build real-life applications with it. At the end of the semester, I'd like to be skilled enough to build a simple AI agent that works with data.

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I hope this class provides fresh lens into the core AI concepts such as search, reasoning, and learning, while exposing me to different strategies and algorithms. Hopefully, that can inspire future studies of mine and help me be more thoughtful and strategic in my applications of AI.

z_last_digits: 572495

student_writeup: I expect to gain a solid and structured understanding of core artificial intelligence concepts through this course. I want to learn how

intelligent agents work and how problems can be solved efficiently using different search strategies and game playing techniques. I am also interested in developing strong knowledge of knowledge representation and reasoning, especially using propositional and first-order logic. Another important area I want to learn is handling uncertainty through probability theory and Bayesian networks. In addition, I aim to strengthen my understanding of learning from examples, including neural networks and reinforcement learning methods such as Markov Decision Processes and Q-learning. Overall, this course will help me improve my analytical thinking, problem-solving ability, and practical AI skills that are useful for real world applications and future research.

z_last_digits: 593961

student_writeup: From CAP 6635 (Artificial Intelligence), I look forward to gaining in-depth knowledge about fundamental concepts in AI and how they can be used in dealing with practical problems. Specifically, I am interested in learning about intelligent agents, problem-solving strategies, search methodologies, knowledge representation systems, reasoning processes, and basic machine learning. Additionally, I look forward to grasping the decision-making processes used by AI systems in uncertain environments and the relative efficiency and effectiveness of varying approaches used in AI systems. This course will therefore provide me with a firm theoretical background in AI on which advanced concepts can be built for my graduate studies.

z_last_digits: 242932

student_writeup: The main goal of this class for me is to learn how non-generative artificial intelligence works generally.

z_last_digits: 573499

student_writeup: I am most interested in Reinforcement Learning, as it appears on the surface to be close to how humans learn. The challenge seems like it would be how to define a reasonable value function to evaluate intermediate states, for example, game moves before you win or lose. I am very curious to learn about this.

I also want to understand how models represent knowledge that can lead to effective reasoning. I hope to have a robust understanding of this so that I can contribute novel research for my PhD.

z_last_digits: 194969

student_writeup: In this course, I hope to learn and expand on knowledge of core concepts in artificial intelligence. I would like to further understand the history and future implications of AI. I want to learn how intelligent agents learn in order to make decisions in AI models. I am especially interested in how AI models are designed and how they utilize information. I would also like to build upon current skills in Python by using. By utilizing current knowledge in Python, I expect to further cultivate these skills through assignments and project. Overall, I want to develop the ability to analyze, design, and evaluate AI systems while understanding their ethical and societal

implications.

z_last_digits: 574085

student_writeup: I eagerly waiting to learn the problem solving by search methods and reinforcement learning techniques.

z_last_digits: 571053

student_writeup: I expect to gain a rigorous, system-level understanding of modern artificial intelligence from core learning theory to practical model design and evaluation. Specifically, I want mastery in representation learning, optimization, and generalization, alongside hands-on experience building, debugging, and deploying ML systems. This course should strengthen my ability to translate real-world problems into mathematically sound AI solutions, integrate models into larger data pipelines, and reason about trade-offs in performance, robustness, and scalability. These skills are essential for my career in applied AI, Geo-AI, and research-driven product engineering. After analyzing the student feedback data, I found that the two most similar answers are:

****z_last_digits: 572638****

student_writeup: "From CAP 6635, I expect to learn the basic ideas behind artificial intelligence, AI concepts like search, knowledge representation, reasoning, and learning. I want to understand how these concepts work in practice and how to analyze simple AI systems. I am also looking forward to gaining experience in designing basic learning systems. Overall, I hope this class gives me a clear and strong foundation in AI that I can build on in future courses and projects."

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These two answers are the most similar because they:

1. ****Share similar goals****: Both students aim to gain a solid understanding of AI concepts, including search, knowledge representation, reasoning, and learning.
2. ****Emphasize practical applications****: Both students want to understand how AI concepts work in practice and how to analyze simple AI systems.
3. ****Highlight the importance of foundational knowledge****: Both students believe that this course should provide a strong foundation in AI, which they can build upon in future courses and projects.
4. ****Use similar language****: Both answers use similar phrases, such as "basic ideas behind artificial intelligence" (z_last_digits: 572638) and "fundamental concepts" (z_last_digits: 593961).

ntal concepts in AI" (z_last_digits: 593961), indicating a similar thought process.

Overall, while there are some minor differences in wording and emphasis, these two answers demonstrate a high degree of similarity in terms of content, goals, and aspirations.

> Finished chain.

RESULT 1:

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Query 2: Summarize Main Topics and Top 5 Themes

```
In [ ]: input_2 = "Using the complete student feedback data, provide: 1) A summary c

print("\n" + "="*80)
print("QUERY 2: Summary of Main Topics and Top 5 Themes")
print("="*80)

result_2 = agent_executor.invoke({"input": input_2})
answer_2 = result_2['output']

print("\n" + "-"*80)
print("RESULT 2:")
print("-"*80)
print(answer_2)
```

```
=====
=====
QUERY 2: Summary of Main Topics and Top 5 Themes
=====
=====
```

> Entering new AgentExecutor chain...

Invoking: `get\_all\_feedback` with `{}`

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z_last_digits: 572495

student_writeup: I expect to gain a solid and structured understanding of core artificial intelligence concepts through this course. I want to learn how

intelligent agents work and how problems can be solved efficiently using different search strategies and game playing techniques. I am also interested in developing strong knowledge of knowledge representation and reasoning, especially using propositional and first-order logic. Another important area I want to learn is handling uncertainty through probability theory and Bayesian networks. In addition, I aim to strengthen my understanding of learning from examples, including neural networks and reinforcement learning methods such as Markov Decision Processes and Q-learning. Overall, this course will help me improve my analytical thinking, problem-solving ability, and practical AI skills that are useful for real world applications and future research.

z_last_digits: 593961

student_writeup: From CAP 6635 (Artificial Intelligence), I look forward to gaining in-depth knowledge about fundamental concepts in AI and how they can be used in dealing with practical problems. Specifically, I am interested in learning about intelligent agents, problem-solving strategies, search methodologies, knowledge representation systems, reasoning processes, and basic machine learning. Additionally, I look forward to grasping the decision-making processes used by AI systems in uncertain environments and the relative efficiency and effectiveness of varying approaches used in AI systems. This course will therefore provide me with a firm theoretical background in AI on which advanced concepts can be built for my graduate studies.

z_last_digits: 242932

student_writeup: The main goal of this class for me is to learn how non-generative artificial intelligence works generally.

z_last_digits: 573499

student_writeup: I am most interested in Reinforcement Learning, as it appears on the surface to be close to how humans learn. The challenge seems like it would be how to define a reasonable value function to evaluate intermediate states, for example, game moves before you win or lose. I am very curious to learn about this.

I also want to understand how models represent knowledge that can lead to effective reasoning. I hope to have a robust understanding of this so that I can contribute novel research for my PhD.

z_last_digits: 194969

student_writeup: In this course, I hope to learn and expand on knowledge of core concepts in artificial intelligence. I would like to further understand the history and future implications of AI. I want to learn how intelligent agents learn in order to make decisions in AI models. I am especially interested in how AI models are designed and how they utilize information. I would also like to build upon current skills in Python by using. By utilizing current knowledge in Python, I expect to further cultivate these skills through assignments and project. Overall, I want to develop the ability to analyze, design, and evaluate AI systems while understanding their ethical and societal

implications.

z_last_digits: 574085

student_writeup: I eagerly waiting to learn the problem solving by search methods and reinforcement learning techniques.

z_last_digits: 571053

student_writeup: I expect to gain a rigorous, system-level understanding of modern artificial intelligence from core learning theory to practical model design and evaluation. Specifically, I want mastery in representation learning, optimization, and generalization, alongside hands-on experience building, debugging, and deploying ML systems. This course should strengthen my ability to translate real-world problems into mathematically sound AI solutions, integrate models into larger data pipelines, and reason about trade-offs in performance, robustness, and scalability. These skills are essential for my career in applied AI, Geo-AI, and research-driven product engineering. Based on the complete student feedback data, here's a summary of the main topics and interests students have about this course:

****Summary:****

Students are excited to learn about various aspects of Artificial Intelligence (AI) and its applications. They want to gain a solid foundation in AI concepts, including intelligent agents, problem-solving, search strategies, knowledge representation, reasoning, and machine learning. Students are also interested in understanding the ethical and societal implications of AI.

****Top 5 Most Common Themes/Topics:****

1. ****Foundational AI Concepts****
2. ****Machine Learning and Neural Networks****
3. ****Intelligent Agents and Problem-Solving****
4. ****Knowledge Representation and Reasoning****
5. ****Applications and Ethics of AI****

****Brief Explanation of Each Topic:****

1. ****Foundational AI Concepts****: Students want to learn the basic ideas behind AI, including its history, core concepts, and future implications. They aim to understand the fundamental principles of AI and how it can be applied to real-world problems.
2. ****Machine Learning and Neural Networks****: Students are interested in learning about machine learning, neural networks, and deep learning. They want to understand how these techniques work, their applications, and how to implement them.
3. ****Intelligent Agents and Problem-Solving****: Students want to learn about intelligent agents, problem-solving strategies, and search methodologies. They aim to understand how AI systems can be designed to make decisions and solve complex problems.
4. ****Knowledge Representation and Reasoning****: Students are interested in learning about knowledge representation, reasoning, and uncertainty. They want to understand how AI systems can represent knowledge, make decisions, and handle uncertainty.

5. **Applications and Ethics of AI**: Students want to learn about the applications of AI in various domains, such as healthcare, and understand the ethical and societal implications of AI. They aim to develop a thoughtful and ethical understanding of AI's role in society.

Overall, students are eager to gain a comprehensive understanding of AI concepts, techniques, and applications, as well as develop practical skills in designing and evaluating AI systems.

> Finished chain.

RESULT 2:

Based on the complete student feedback data, here's a summary of the main topics and interests students have about this course:

Summary:

Students are excited to learn about various aspects of Artificial Intelligence (AI) and its applications. They want to gain a solid foundation in AI concepts, including intelligent agents, problem-solving, search strategies, knowledge representation, reasoning, and machine learning. Students are also interested in understanding the ethical and societal implications of AI.

Top 5 Most Common Themes/Topics:

1. **Foundational AI Concepts**
2. **Machine Learning and Neural Networks**
3. **Intelligent Agents and Problem-Solving**
4. **Knowledge Representation and Reasoning**
5. **Applications and Ethics of AI**

Brief Explanation of Each Topic:

1. **Foundational AI Concepts**: Students want to learn the basic ideas behind AI, including its history, core concepts, and future implications. They aim to understand the fundamental principles of AI and how it can be applied to real-world problems.
2. **Machine Learning and Neural Networks**: Students are interested in learning about machine learning, neural networks, and deep learning. They want to understand how these techniques work, their applications, and how to implement them.
3. **Intelligent Agents and Problem-Solving**: Students want to learn about intelligent agents, problem-solving strategies, and search methodologies. They aim to understand how AI systems can be designed to make decisions and solve complex problems.
4. **Knowledge Representation and Reasoning**: Students are interested in learning about knowledge representation, reasoning, and uncertainty. They want to understand how AI systems can represent knowledge, make decisions, and handle uncertainty.
5. **Applications and Ethics of AI**: Students want to learn about the applications of AI in various domains, such as healthcare, and understand the ethical and societal implications of AI. They aim to develop a thoughtful and ethical understanding of AI's role in society.

Overall, students are eager to gain a comprehensive understanding of AI concepts, techniques, and applications, as well as develop practical skills in designing and evaluating AI systems.

Summary

Both queries have been completed. The first query identified the most similar student answers, while the second query provided a comprehensive analysis of main topics and the top 5 themes students are interested in.